

**Transportation Access and Potentially Preventable Emergency Room
Visits: Case Study of the Second Avenue Subway Expansion in New
York City**

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Abstract

Under Andersen's (1995) Behavioral Model of Health Services Use, healthcare utilization and outcomes are shaped by environmental conditions, population characteristics, and the availability of enabling resources. However, the relative contribution of these factors may vary across populations and contexts. This thesis asks whether increased subway access following the 2017 Second Avenue Subway expansion affected healthcare outcomes in local and connected ZIP codes, hypothesizing that improved transportation access to healthcare via the Q line, as one enabling resource among many, would reduce potentially preventable emergency department visits (PPV) by facilitating access to routine medical care for residents of neighborhoods served by the Q train. Demographic, emergency department, and spatial data were drawn from the American Community Survey (ACS), Statewide Planning and Research Cooperative System (SPARCS), and Metropolitan Transportation Authority (MTA). Both studies use ZIP code-year panel data and difference-in-differences/event-study approaches, although the strength of causal interpretation differs between the two analyses. Study 1 evaluates the local effect of the Q-line extension on PPV rates in directly affected Upper East Side ZIP codes relative to selected Manhattan controls with comparable pre-treatment PPV rates. The estimated post-expansion effect is positive but statistically nonsignificant, contrary to the hypothesis. Study 2 examines patterns across the broader Q-line corridor, incorporating variation in neighborhood change and proximity to the Q line. Results are largely null or mixed across specifications and subgroup analyses. Uncertainty regarding the parallel trends assumption and potential post-treatment bias introduced by the propensity-score matching procedure mean that Study 2 should be interpreted as exploratory rather than causal. Taken together, the findings do not provide evidence that increased subway access reduced PPV rates, but they also do not establish that the expansion increased PPV or had no effect. The results suggest that transportation access may operate as only one of several interacting enabling factors shaping healthcare utilization and outcomes.

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Introduction

Phase 1 of the Second Avenue Subway Expansion in New York City was a long time coming; while the project was first proposed in 1929, it was not until 2017 that the Second Avenue Subway line opened to the public, greatly expanding Q-train access to the Upper East Side (MTA, 2025). Despite its wealth and density, this area has historically had quite limited subway access, isolating it from the rest of the city; the MTA estimates that the 2017 expansion provides service to nearly 200,000 riders daily, reduces overcrowding on other train lines by 40% on average, shortens travel time, and enables easier access to the Upper East Side for Brooklyn and Queens residents (MTA, 2025). Further expansion of the Second Avenue Line is currently in progress, and is planned to expand Q line access further uptown into Harlem (MTA, 2025).

Currently, the upper east side is home to highly ranked outpatient medical centers such as New York Presbyterian Weill-Cornell and Medical Offices of Manhattan - Upper East Side (Google, n.d.). It is well documented that transportation may be a barrier to health care access, especially in disadvantaged communities (Syed, Gerber, and Sharp, 2013; Rask et al., 1994). Similarly, Andersen (1968)'s widely used behavioral model of health utilization (as cited and later revisited in Andersen 1995) posits that health service utilization can be determined by three factors including individual factors, enabling factors that allow individuals to access care, including transportation, and need factors, such as the level of perceived need for health services (Andersen, 1995). The 2017 expansion of access to the upper east side presents an opportunity to study the impact of increased access to convenient transportation on health outcomes using a natural quasi-experiment.

This thesis asks whether increased subway access via the 2017 Second Avenue Expansion affected healthcare outcomes in local and connected ZIP codes. I propose that increased access to healthcare via the second avenue Q-line, as one enabling factor among many, will improve healthcare outcomes by way of increased health service utilization for residents of neighborhoods served by the

Q-train. I examine this question using risk-adjusted potentially preventable emergency department visit rates as an indicator of healthcare access and outcomes, employing a dynamic difference-in-differences model. Study 1 examines the impact of the 2nd Avenue expansion on local (Upper East Side) ZIP codes, finding a positive but statistically nonsignificant estimate, in opposition to the hypothesis that improved transportation access would improve medical outcomes. Study 2 explores the expansion's impact on healthcare outcomes in Q-line ZIP codes outside of the Upper East Side, finding largely mixed or null results. Finally, assumptions and limitations are discussed at the end of this paper.

Literature Review

Under Andersen's (1995) Behavioral Model of Health Services Use, outcomes are influenced by environment, population characteristics including individual traits and the availability of "enabling resources" (e.g., resources that allow an individual to seek health care, including access to transportation), and health behaviors (e.g., personal health practices including exercise and diet, and use of health services). This model is recursive, such that an individual's health status can further impact their health behaviors and other predisposing factors, including access to and use of "enabling resources".

Thus, Andersen (1995) suggests that behaviors such as walking to and from the subway and access to direct transportation are both elements of a dynamic system that feed into one another, ultimately determining healthcare utilization and health outcomes. The question remains, however, how much each aspect of the model contributes to utilization and outcomes for different population groups.

On Routine and Preventative Medical Care

Access to routine medical care is essential for maintaining health, early detection of serious illness, and management of chronic and pre-existing conditions. For example, Schechtman et al. (2008) demonstrate that adherence to medical appointments is a predictor of metabolic (disease) control in patients with diabetes. Moreover, they found that racial disparities in metabolic control were diminished when they controlled for health insurance status and the missed appointment rate, further emphasizing the need for routine care in vulnerable populations.

Despite the tangible benefits of routine care, many patients face significant barriers to access. Wolfe et al. (2020) found that 5.8 million Americans delayed medical care in 2017 due to transportation barriers (e.g., lack of transportation); transportation barriers were especially burdensome for already disadvantaged groups such as those of Hispanic descent and those with chronic or pre-existing conditions. In other words, disadvantages “stack” atop each other, in line with Andersen (1995)’s Behavioral Model of Health Services Use. The more disadvantaged you are, the more barriers you face, and the more barriers you face, the more disadvantaged you become.

Access as a mechanism for improving healthcare utilization and disparities across groups

Studies examining the removal of transportation barriers have consistently found that improving access increases healthcare utilization. Smith et al. (2022) studied the impact of a new light rail line opening on missed appointments (not including cancelled appointments) at clinics in the area using a difference-in-differences experimental design. They found that patients living near the new light rail line after its construction were less likely to skip appointments compared to those living farther away. The results were especially pronounced for low-income patients on Medicaid (Smith et al., 2022). Convenience may be particularly linked to increased healthcare utilization, which should theoretically link to clinical outcomes in turn. Similarly, Im et al. (2025) found that respondents in

Atlanta, a major metropolitan area, frequently cited public transportation fare costs and inconvenient public transit (e.g., infrequent train schedules and long wait times) as salient barriers to access; inconvenient public transit was even sometimes cited as a reason for missed appointments, further disadvantaging low-income respondents through fees in addition to transportation and opportunity costs, such as time spent waiting between appointments (Im et al., 2025).

Finally, in a meta-analysis of studies on transportation barriers to health care access, Syed, Gerber, and Sharp (2013) found that reliable access to transportation was consistently linked to increased healthcare utilization across studies. One study in Atlanta found that the use of public transportation or walking was a predictor of lacking a regular source of medical care for low-income adults. It was also associated with a higher likelihood of delaying care for this population (Rask et al., 1994, as cited in Syed, Gerber, and Sharp, 2013). Though the majority of research on transportation barriers has historically focused on rural and car-centered areas, transportation-related barriers may still delay patient utilization of care among vulnerable populations, even in areas that do have usable public transportation (Syed, Gerber, & Sharp, 2013).

Telehealth and Associated Tradeoffs

Telehealth (virtual appointments) offers an intuitive solution to concerns about transportation barriers, yet it is still unclear how effective telehealth is compared with traditional in-person appointments, and whether it is equally effective across medical needs and appointment types. A recent meta-analysis on telehealth efficacy during the COVID-19 pandemic found mixed results; while there was evidence supporting a lower rate of missed appointments (e.g., increased utilization), findings also included lower rates of up-to-date labs amongst patients receiving care via telehealth (Hatef et al., 2024). Thus, switching from in-person appointments to telehealth involves tradeoffs that may or may not be acceptable for a given case. Despite the value that telehealth can provide, it may also allow patients to fall behind on laboratory testing, ultimately delaying care.

Mixed evidence

While this access-to-outcomes mechanism for health outcomes has been studied extensively, findings have been mixed and have often relied on self-report measures. Solomon et al. (2020) performed a meta-analysis of 8 studies on the impact of transportation-based interventions on various health outcomes. They found that all studies used measures of healthcare utilization, while only one studied direct health outcomes, and a great deal of the peer-reviewed research on the topic was over a decade old. Moreover, the sole study that measured clinical health outcomes did not find a meaningful difference between patients who received transportation assistance and those who did not, though the authors hypothesize this may be due to concurrent barriers such as housing or food insecurity, or the severity of pre-existing conditions that may mediate the relationship between transportation and clinical outcomes (Solomon et al., 2020). The link between healthcare utilization and health outcomes is not as clear-cut as one might believe, leaving a clear need for new peer-reviewed research that models the relationship between transportation barriers, clinical outcomes, and other mediating factors. Furthermore, many studies estimate travel burden using time and distance, which are often difficult to capture accurately and consistently due to daily changes in traffic and road outages (Kabayundo et al., 2025).

This study differs by using the number of train stops as a standardized distance measure, reducing the need to estimate terrain features and roads, which is very difficult to capture accurately. Despite New York City's extensive transportation network, there are well-documented disparities in accessibility which should be studied (Baghestani et al., 2024). The present study examines the question: to what extent do transportation barriers to healthcare exist in New York City, and what impact do they have on clinical patient outcomes?

Theory and Hypothesis

We propose a conceptual framework in which the 2017 subway expansion increases transportation accessibility, which may contribute to improved health outcomes by increasing preventive care utilization among vulnerable populations. Under the Andersen (1995) recursive model, transportation access functions as a key enabling factor that shapes individuals' ability to access routine health care, rather than as a direct determinant of health outcomes. Due to the recursive nature of this model, personal health practices, enabling resources, and personal characteristics may jointly influence health outcomes, which may feed back to reshape behaviors and outcomes over time.

Under this framework, increased transportation accessibility may contribute to improved health outcomes via a healthcare utilization pathway, through which a more direct mode of transportation interacts with other predisposing factors to increase use of health services by reducing the time and effort needed to obtain routine care. This improved accessibility is expected to interact with individuals' predisposing characteristics and enabling resources to facilitate greater use of preventive and routine health services. We hypothesize that risk-adjusted potentially preventable emergency department visit rates will decline in affected areas following the expansion of the Q-line. Because neighboring ZIP codes do not experience comparable improvements in transit accessibility, we expect any observed reductions in potentially preventable emergency department visit rates to be smaller or absent in those areas.

Method

To study the relationship between transportation access and healthcare utilization using quasi-experimental methods, we implement two complementary studies: Study 1 evaluates the impact of the Q line extension on preventable emergency department visits in the intervention area. We

compare hospitalizations on the Upper East Side to matched Manhattan control ZIP codes. However, given that the Upper East Side is already a high-income, transit-rich area, this design may not be well-suited to detect access-driven health effects.

To address this limitation, Study 2 expands the analysis to a broader set of ZIP codes and incorporates variation in neighborhood change (gentrification) and proximity to the Q line corridor. Both studies use panel data at the ZIP code - year level and use difference-in-differences and event study designs to estimate effects of the Q-line extension on potentially preventable emergency room visits.

DiD

Difference-in-differences estimation (DiD) uses a regression model to estimate the causal effect of an intervention by using longitudinal data to construct a counterfactual. A dynamic DiD, or event study, estimates effects at different times relative to treatment, whereas a conventional post-treatment indicator summarizes an average post-treatment effect. An event study can describe dynamic patterns and assess pre-treatment differences, but it does not automatically reduce bias or resolve problems associated with staggered treatment adoption (LOSTstats).

Assumptions

DiD relies on several core assumptions. Most critical is the assumption of parallel trends, which requires that, absent treatment, average outcomes in the treated and control groups would have followed parallel changes over time. Violation of the parallel trends assumption may lead to significant error, and makes results unreliable (Columbia Mailman School). Identification also requires a credible comparison group, no anticipatory outcome responses in the designated pre-treatment period, and an exposure definition that addresses spillovers between treated and comparison units (Columbia Mailman School).

This project assumes that residents of neighborhoods along the Q line had parallel changes in health outcomes as other neighborhoods of equivalent socioeconomic status and racial composition, and that these parallel trends would have continued without the opening of the expansion project; pre-treatment outcome trajectories can inform the plausibility of this assumption but cannot confirm it (LOSTstats).

This project also assumes that residents of NYC neighborhoods did not self-select into treatment and control neighborhoods (ie. neighborhoods on and off the Q line) based on preference for access to healthcare or transportation. I further assume that there are no other policy related interventions or demographic shifts in the years following the intervention that may confound the results. Violation of these assumptions compromises the credibility of my findings.

Study 1: Upper East Side Q-Line Expansion

Data

The primary dataset comes from the New York State SPARCS database, which provides ZIP code-level measures of hospital emergency department utilization. The outcome variable is the risk-adjusted rate of potentially preventable emergency department visits (PPV) per 100 residents. Risk-adjusted rates are used to account for demographic variation across ZIP codes, including age, gender, and race/ethnicity. The analytic sample spans 2011–2022 and is structured as a ZIP code-by-year panel. Observations with missing outcome values were excluded. Missingness was concentrated outside New York City and did not substantively affect the Manhattan sample. One Upper East Side ZIP code (10075) was missing across all years and therefore excluded from analysis.

Treatment and Control Groups Selection

The treatment group consists of three Upper East Side ZIP codes directly affected by the 2017 Q-line expansion: 10021, 10028, and 10128. The control group includes Manhattan ZIP codes not directly affected by the Q-line expansion. Restricting the control group to Manhattan ensures comparability in healthcare access, urban density, and baseline infrastructure.

Nine control ZIP codes were selected from a pool of Manhattan ZIP codes based on similarity in pre-treatment PPV rates (2011–2016). I calculated mean baseline PPV rates for each ZIP code and retained those within a comparable range to the treated group. Alternative specifications (including matching on pre-treatment trends) were explored, but broader control groups produced more stable and consistent pre-treatment trends and were therefore used in the primary analysis.

Roosevelt Island (ZIP code 10044) was included in the control group. Although geographically proximate to treated areas, it was not directly served by the subway expansion. Including this ZIP code improved alignment in pre-treatment trends, and results are robust to its exclusion.

Study 1 Figure 1. Baseline PPV Distribution Across Manhattan ZIP Codes

Dashed reference lines mark treated Upper East Side ZIP codes

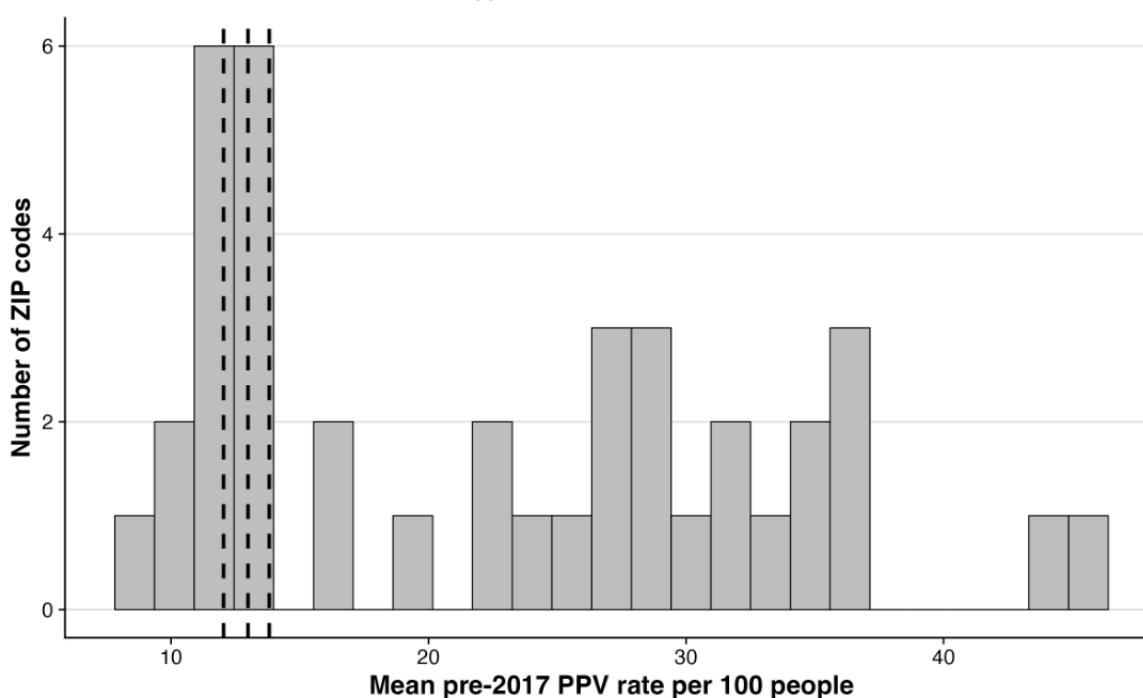


Figure 1 shows the mean PPV rate for the three treatment ZIPs (orange) against all Manhattan ZIP codes. Manhattan ZIPs had a wide range of mean PPV values and a right skew, with treatment ZIPs falling towards the lower half of the range (albeit, above the overall average due to the skew). Table 1 similarly shows slightly higher mean PPV and smaller range for treatment ZIPs compared to control ZIPs or Manhattan overall.

Study 1 Table 2. PPV Summary Statistics

Group	Mean PPV	SD PPV	Minimum PPV	Maximum PPV	n	group
Overall	10.79	2.55	5.57	22.39	156.00	
Control	10.56	2.57	5.57	22.39	117.00	Control
Treated	11.49	2.36	6.97	16.67	39.00	Treated

Comment: This table summarizes PPV rate levels in the Study 1 analytic sample overall and by treatment group.

Table 1 summarizes PPV rate levels in the Study 1 analytic sample overall and by treatment group.

Parallel Trends and Model Validation

The parallel trends assumption was evaluated using both graphical inspection and formal pre-treatment tests. Trends in PPV rates for treatment and control ZIP codes were compared over the pre-treatment period (2011–2016). Multiple control group constructions were tested, including restricting to tighter baseline ranges and matching on pre-treatment slopes. While tighter restrictions improved similarity in levels, broader control groups yielded more stable and parallel trends over time and were therefore selected for the main specification.

Robustness Checks

To account for potential confounding effects of the COVID-19 pandemic, all models were re-estimated excluding observations from 2020 onward. This allows for comparison of estimates in a period unaffected by pandemic-related disruptions to healthcare utilization.

Difference-in-Differences Models

The primary specification is a two-way fixed effects difference-in-differences model:

$$Y_{zt} = \beta(Treated_z \times Post_t) + \alpha_z + \delta_t + \varepsilon_{zt}$$

where Y_{zt} is the risk-adjusted PPV rate in ZIP code z during year t ; $Treated_z$ indicates Upper East Side treatment ZIP codes; $Post_t$ indicates observations from 2017 onward; α_z are ZIP code fixed effects; δ_t are year fixed effects; and ε_{zt} is the error term.

Standard errors are clustered at the ZIP code level. This specification estimates the average effect of increased subway access on preventable emergency department visits.

Event Study Specification

To assess dynamic treatment effects and evaluate the parallel trends assumption, I estimate an event study model:

$$Y_{zt} = \sum_{k \neq -1} \beta_k [Treated_z \times 1(event_time_t = k)] + \alpha_z + \delta_t + \varepsilon_{zt}$$

where the β_k coefficients represent treatment effects at each event time relative to the omitted pre-treatment year.

Sun–Abraham Estimator

To address potential bias in two-way fixed effects models under treatment effect heterogeneity, I estimate a Sun and Abraham (2021) specification as a robustness check. This approach produces cohort-specific event-time estimates and avoids the negative weighting issues associated with standard DiD estimators.

Study 2: Impact of Expansion down the Q line (Increased Access impact on Potentially Preventable Emergency Department Visits)

Study 1 focuses on a high-income, well-served area of New York City where transportation access may not be a binding constraint on healthcare utilization. Study 2 extends the analysis to a broader set of ZIP codes to evaluate whether transportation effects are more pronounced in areas with greater structural barriers to care, and uses demographic data from the American Community Survey (ACS) for controls and matching.

Data and Sample

Study 2 uses the same Q-train station dataset, SPARCS dataset and outcome variable as Study 1, but expands the geographic scope beyond Manhattan's Upper East Side to Brooklyn and lower Manhattan. ZIP codes are selected to capture variation in baseline healthcare access, socioeconomic conditions, and proximity to transit infrastructure. I queried ACS Demographic data for the years 2011 through 2022 using the tidycensus R package for reproducibility. In addition, a MODZCTA crosswalk from NYOpenData was used to convert the ACS's ZCTA-based observations to ZIP codes in line with the SPARCS and Q-line datasets.

Treatment in Study 2 is defined based on exposure to the Q-line. A 1.5-km buffer around Q-line stations initially identified 35 candidate corridor ZIP codes. Five ZIP codes directly exposed to the new Upper East Side stations (10021, 10028, 10029, 10065, and 10128) were excluded to separate direct exposure from broader corridor access, leaving 30 treated ZIP codes eligible for matching. 30 controls were selected using propensity-score matching, and three matched ZIP codes (treated ZIP 10075 and control ZIPs 10282 and 11109) had no usable PPV observations in the final analytic panel. This yielded a final Study 2 outcome sample of 29 treated and 28 control ZIP codes (57 total).

ACS Variables

ACS data are used to construct a gentrification-adjusted matching dataset, spatial data define treatment exposure, matched ZIP codes are merged into a panel of hospital outcomes, and effect sizes are estimated using event-study difference-in-differences models. We created several derived variables, including percent of Black, Hispanic, and White residents, poverty rate, college educated

rate, and a gentrification score for each ZIP. See appendix for complete list of variables used and created.

ZIP code and ZCTA matching

The ACS and SPARCS datasets used two different measure of spatial area: the ACS uses Zip Code Tabulation Areas (ZCTA) to define geographic polygons used for mapping and spatial analysis (Census.gov, 2023), while the SPARCS dataset uses patient ZIP code, which are point based. I used the Modified Zip Code Tabulation Areas shapefile from NYC OpenData (https://data.cityofnewyork.us/Health/Modified-Zip-Code-Tabulation-Areas-MODZCTA-/pri4-ifjk/about_data,2020), and performed a left join to the Q-line data (ZIP code level). To merge these datasets into a dataframe with a single ZCTA (which are called ZIPs henceforth) I identified ZIP codes present in the spatial dataset but missing from the ACS data by computing the set difference between the two ZIP code lists, and then grouped the data by ZCTA to create a workable dataframe with one ZIP code in each row.

NAs and Missingness

Data were inspected for missingness, and evaluated for geographic and temporal clustering, as well as systematic differences. Observations were counted by year and ZIP. All ZIPs with a missing value anywhere also had a missing income column (see table 1). Of 2,508 rows, 381 (15.2%) had at least one missing value and 2,127 had none. The map suggests that some missing observations correspond to parks and waterways, but the consequences of excluding populated areas require further assessment (Figure 1).

Study 2 Table 1. Mean Values for ZIPs with Missingness Compared to ZIPs Without

Group	Mean income	Mean college educated share	Mean pct Black	n
ZIPs without missingness	72,598.60		19.88	2,127.00
ZIPs with missingness		0.22	34.25	381.00

Comment: This table compares ACS-derived ZIP-year rows with any missingness to rows without missingness using the same logic as the missingness diagnostic chunk in the thesis draft.

Table 1. Compares mean values for ZIPs with a missing value in any column to ZIPs with no missing values

Defining Spatial Exposure

To define Q-line exposure (which ZIP codes should be considered “exposed”, I loaded subway station data and converted it into spatial features; Q-line stations were then identified and isolated. A 1 km buffer was created around new Upper East Side stations to define direct exposure ZIPs, and a broader 1.5 km buffer was created around all other Q-line stations to define a corridor of exposure. ZIP codes intersecting these buffers were extracted as exposure candidates (eg. treatment ZIPs).

Measuring Gentrification

To define gentrification in each ZIP code before (2011–2016) and after treatment (2017–2022), I computed average income, education, and poverty over time. Citywide average changes were computed as a benchmark, and ZIP-level changes were then normalized relative to the NYC average. I then calculated a composite gentrification index by standardizing and combining relative income, education, and poverty changes.

Propensity Score Matching

After constructing a ZIP-level dataset including treatment status, income (pre and post), and gentrification index and observations with missing covariates were removed, I used the R package MatchIT to perform nearest-neighbor 1:1 matching using income (pre and post) and the gentrification

index. Balance diagnostics were evaluated before and after matching, and a matched sample with equal numbers of treated and control ZIPs was obtained. The table below shows summary statistics across treated and control groups before and after matching; nearest-neighbor matching significantly improved similarity across variables. Matched treatment and control ZIPs were then extracted for analysis.

An important limitation here is that the matching procedure incorporates income and neighborhood changes measured after the subway expansion. Because these characteristics may themselves have been influenced by the expansion, the resulting matched groups may not provide a clean basis for causal comparison. Study 2 is therefore interpreted as exploratory rather than as supporting firm causal conclusions.

Study 2 Table 2. Matching Balance Before and After Nearest-Neighbor Matching

sample	group	n	Pre-period income	Post-period income	Gentrification index
After matching	Control	30.00	76,802.27	98,283.62	0.99
After matching	Treated	30.00	79,122.31	102,211.47	0.82
Before matching	Control	142.00	60,509.86	74,618.96	-0.13
Before matching	Treated	30.00	79,122.31	102,211.47	0.82

Comment: This table checks whether treated and control ZIPs look more comparable after matching on income and gentrification measures.

Study 2 Table 2: This table checks whether treatment and control ZIPs look more comparable after matching on income and gentrification measures.

Parallel Trends, Heterogeneity, and DiD Models

As in study 1, The parallel trends assumption was evaluated using both graphical inspection and formal pre-treatment tests. Multiple control group constructions were tested, including restricting to tighter baseline ranges and matching on pre-treatment slopes. Because of suspected heterogeneity

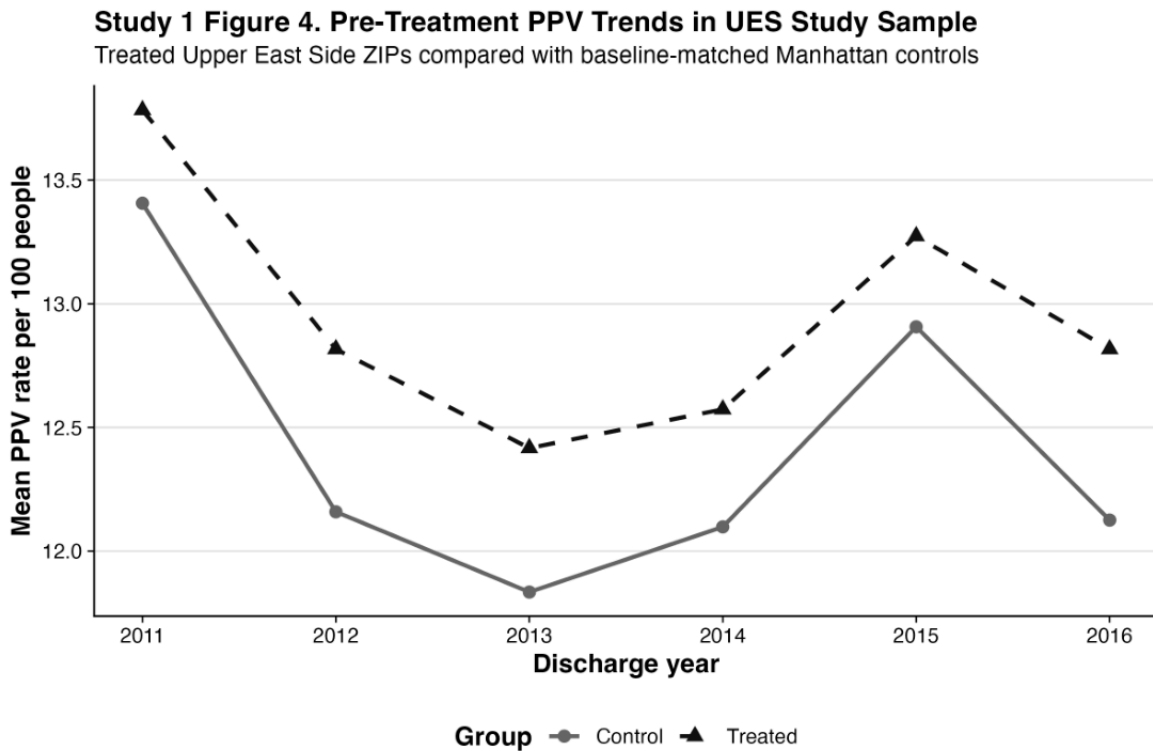
of the treatment zips, I created income and distance bins to categorize ZIPs as either “high”, “medium”, and “low” distance or income. Parallel trends were tested by income and distance bins. Similarly, treatment effects were estimated using standard DiD, event-study DiD, and Sun-Abraham Estimator. Effects were also estimated for each income and distance bin.

Results

Study 1

Parallel Trends

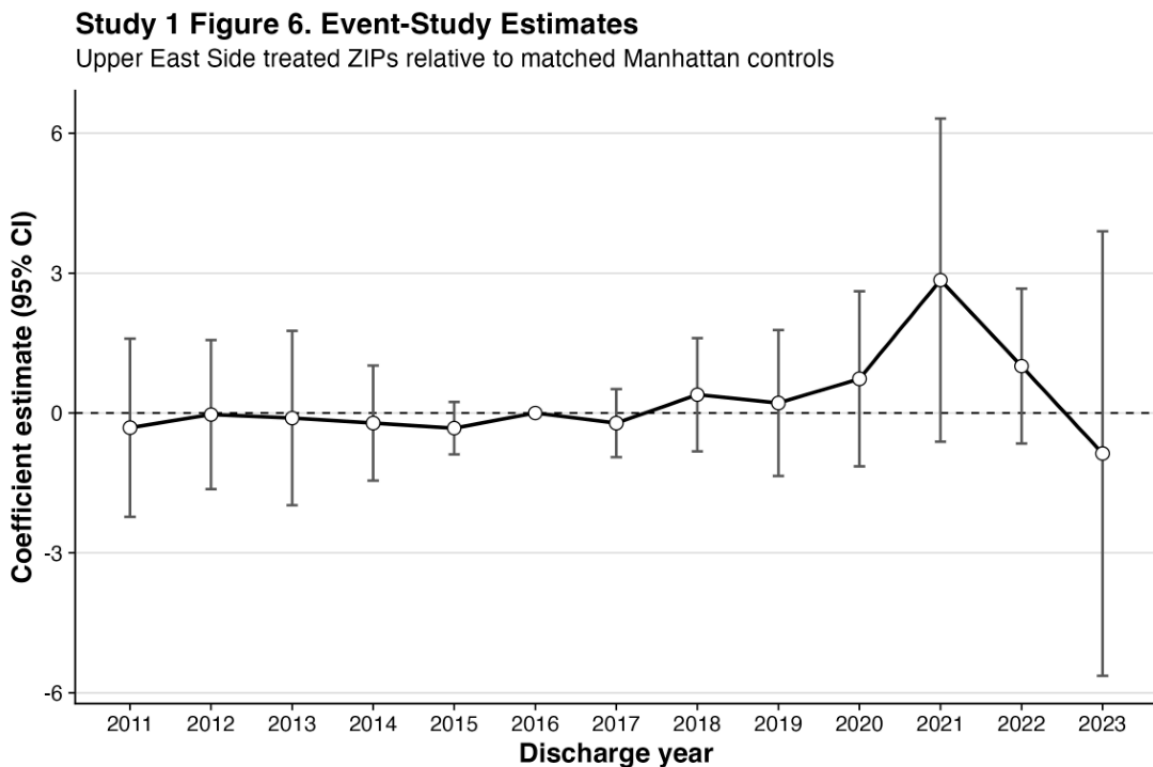
Visual examination and the reported tests did not identify clear evidence of differential pre-treatment trends.



The formal parallel pretrends test did not reveal significant differences between treatment and control slopes in pre-treatment trends (all $p > 0.05$), although failure to reject these tests does not establish parallel trends.

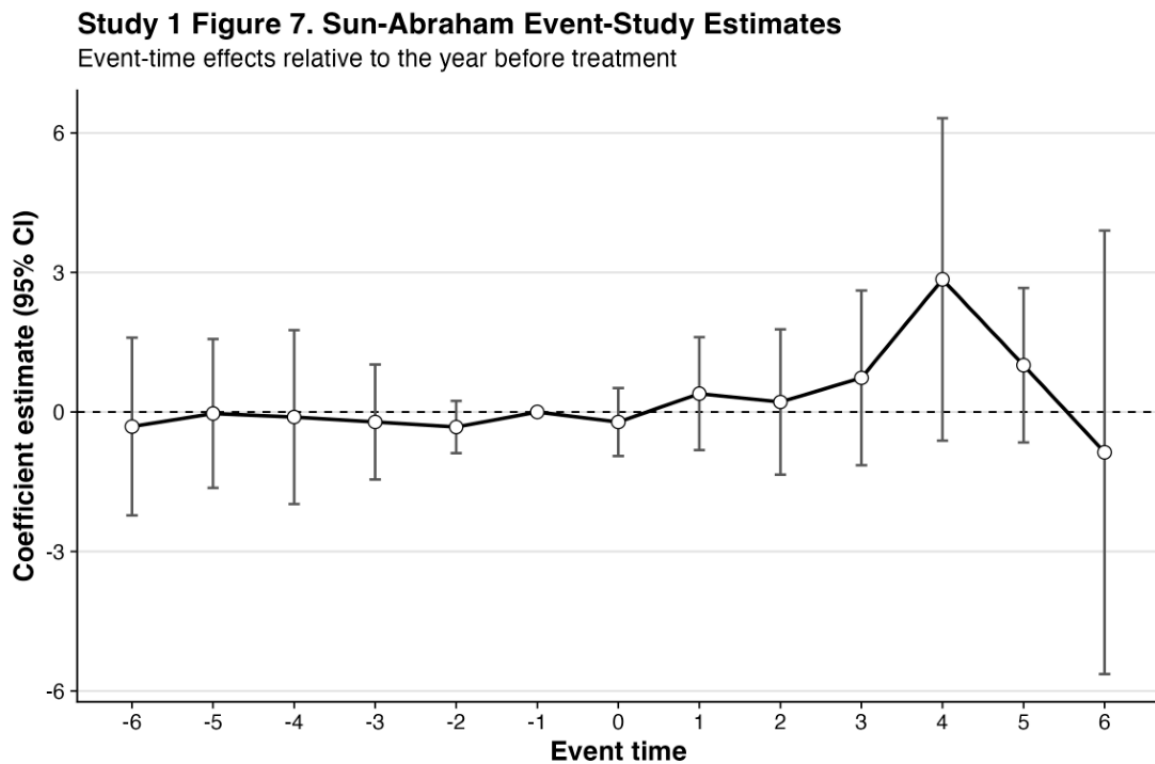
Study 1 Regression Results

At the local level, the Q-line expansion was associated with a 0.75 increase in preventable ED visits per 100 residents per year (95% CI: -1.38 to 2.88), though this effect was not statistically significant ($p = 0.453$). When analyses were restricted to the pre-COVID period (2011–2019), the estimated effect was smaller (0.30 per 100 residents per year, 95% CI: -1.18 to 1.77, $p = 0.666$), although neither specification precisely identifies the direction of the association.



Both specifications show positive point estimates, but their confidence intervals include reductions as well as increases in potentially preventable ED visits. These results therefore do not

establish that the expansion increased visits. Excluding 2020 onward yields a smaller point estimate, but does not establish whether the pandemic explains differences between specifications.



Study 2

Parallel Trends

Visual inspection of pre-treatment trends (2011–2016) indicates that treatment and control ZIP codes generally move in the same direction over time; however, the magnitude of changes differs across groups. While this suggests some shared underlying dynamics, the lack of consistent alignment in levels and slopes raises concerns about the strength of the parallel trends assumption.

Formal pre-trend tests using an event-study specification further support this conclusion. Although not uniformly statistically significant, several pre-treatment coefficients deviate from zero, indicating that treated and control ZIP codes may have experienced differential trends prior to the intervention. Taken together, these results suggest that the credibility of the parallel trends assumption

is uncertain in the full matched sample. Combined with concerns about post-treatment bias in the Study 2 propensity score matching procedure, these trends reinforce that Study 2 should be interpreted as exploratory, rather than supporting firm causal claims.

Study 2 Table 10. Formal Pre-trend Test

model	term	estimate_display	std error	statistic	p value	conf low	conf high
Overall matched sample	Discharge Year x x 2011 x treated	-1.268*	0.67	-1.89	0.06	-2.61	0.07
Overall matched sample	Discharge Year x x 2012 x treated	-0.430	0.64	-0.67	0.51	-1.71	0.85
Overall matched sample	Discharge Year x x 2013 x treated	-1.156***	0.41	-2.79	0.01	-1.99	-0.33
Overall matched sample	Discharge Year x x 2014 x treated	-0.374	0.29	-1.30	0.20	-0.95	0.20
Overall matched sample	Discharge Year x x 2015 x treated	0.846***	0.26	3.29	0.00	0.33	1.36

Comment: This table reports the formal pre-trend test for the overall matched Study 2 sample before splitting by income or distance. Significance codes: *** p < 0.01, ** p < 0.05, * p < 0.1.

Study 2 table 10: Formal pre-tend test for matched sample and controls

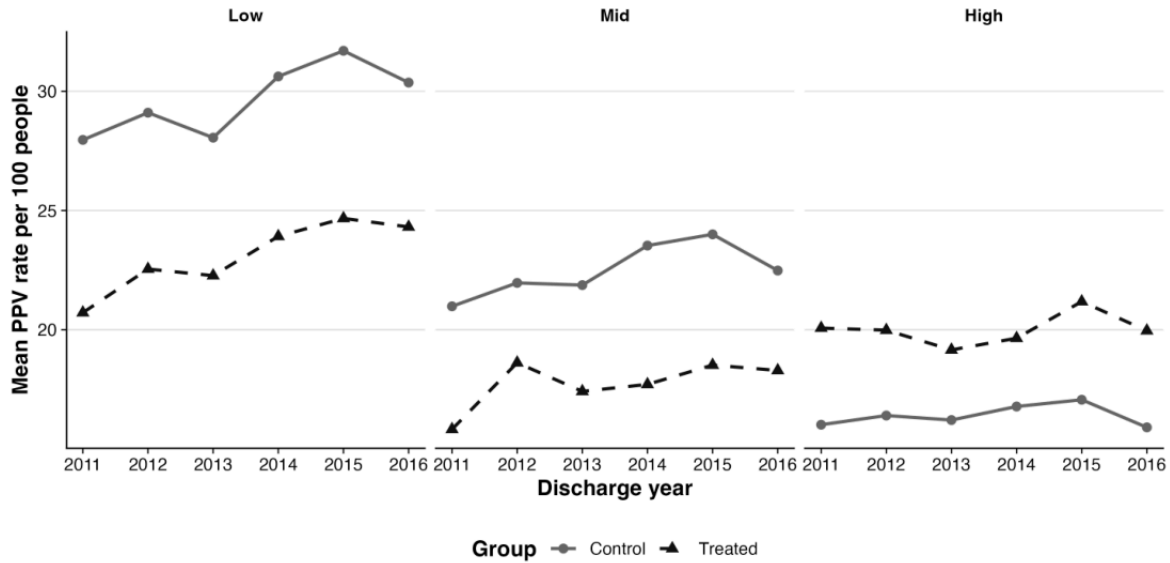
Heterogeneous Parallel Trends

Income

When stratifying by income level, pre-treatment trends exhibit meaningful variation across groups. In higher-income ZIP codes, treated and control units display more similar trajectories, whereas lower-income ZIP codes show greater divergence in both levels and slopes prior to treatment. These differences are also reflected in stratified pre-trend regressions, which indicate that the parallel trends assumption is more plausible for some income groups than others.

High income ZIPs were more plausibly parallel after 2013 upon visual inspection than mid or low income ZIPs, but upon formal testing showed a statistically significant pretreatment coefficient (1.22, $p < .01$), violating the parallel trends assumption. Conversely, neither the low or medium income had any significant coefficient after 2013, despite their visual presentation.

Study 2 Figure 2. Pre-Treatment PPV Trends by Baseline Income Tercile



Study 2 Table 11. Formal Parallel Trends Test by Income Bin

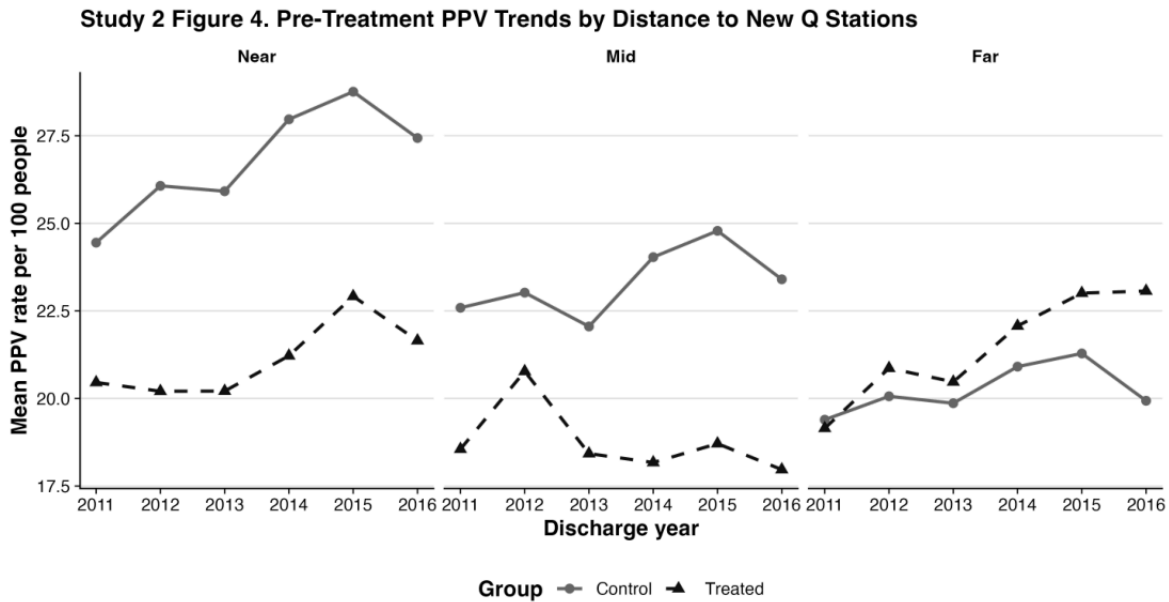
model	term	estimate_display	std error	statistic	p value	conf low	conf high
Low income bin	Discharge Year x x 2011 x treated	-3.596***	0.91	-3.97	0.00	-5.41	-1.78
Low income bin	Discharge Year x x 2012 x treated	-1.768**	0.86	-2.07	0.04	-3.48	-0.05
Low income bin	Discharge Year x x 2013 x treated	-2.044***	0.69	-2.97	0.00	-3.42	-0.66
Low income bin	Discharge Year x x 2014 x treated	-0.395	0.35	-1.14	0.26	-1.09	0.30
Low income bin	Discharge Year x x 2015 x treated	0.357	0.39	0.91	0.37	-0.43	1.14
Mid income bin	Discharge Year x x 2011 x treated	-2.470***	0.80	-3.09	0.00	-4.07	-0.87
Mid income bin	Discharge Year x x 2012 x treated	0.322	0.95	0.34	0.73	-1.57	2.22
Mid income bin	Discharge Year x x 2013 x treated	-0.878	0.73	-1.19	0.24	-2.35	0.59
Mid income bin	Discharge Year x x 2014 x treated	-0.585	0.44	-1.34	0.19	-1.46	0.29
Mid income bin	Discharge Year x x 2015 x treated	0.225	0.43	0.53	0.60	-0.63	1.08
High income bin	Discharge Year x x 2011 x treated	0.111	0.89	0.12	0.90	-1.68	1.90
High income bin	Discharge Year x x 2012 x treated	0.022	0.96	0.02	0.98	-1.89	1.94
High income bin	Discharge Year x x 2013 x treated	-0.804	0.58	-1.40	0.17	-1.96	0.35
High income bin	Discharge Year x x 2014 x treated	-0.315	0.45	-0.70	0.49	-1.22	0.59
High income bin	Discharge Year x x 2015 x treated	1.222***	0.36	3.43	0.00	0.51	1.94

Comment: This table reports separate formal pre-trend models for the low-, mid-, and high-income Study 2 ZIP bins.
Significance codes: *** p < 0.01, ** p < 0.05, * p < 0.1.

Distance

We also assessed parallel trends across distance-based strata relative to the new subway stations. For ZIP codes located farther from the intervention, pre-treatment trends appear more stable

and more closely aligned, particularly after approximately 2013. In contrast, ZIP codes in closer proximity to the new stations exhibit more pronounced divergence in pre-treatment trajectories, suggesting potential violations of the parallel trends assumption in these areas. Formal analysis shows that near and middle-distance ZIPs both had significant coefficients for 2015, raising concerns about the parallel trends assumption. Far away ZIP codes had statistically significant pre-treatment coefficients in 2013 and 2015, but not in 2014.



Formal Parallel Trends Test by Distance Bin

Study 2 Table 12. Formal Parallel Trends Assessment by Distance Bin

model	term	estimate_display	std error	statistic	p value	conf low	conf high
Near distance bin	Discharge Year x x 2011 x treated	-1.195	0.97	-1.24	0.22	-3.13	0.74
Near distance bin	Discharge Year x x 2012 x treated	-1.445**	0.63	-2.29	0.03	-2.71	-0.18
Near distance bin	Discharge Year x x 2013 x treated	-1.441**	0.65	-2.22	0.03	-2.74	-0.14
Near distance bin	Discharge Year x x 2014 x treated	-0.433	0.50	-0.86	0.39	-1.44	0.57
Near distance bin	Discharge Year x x 2015 x treated	1.265***	0.40	3.13	0.00	0.46	2.07
Mid distance bin	Discharge Year x x 2011 x treated	0.584	1.16	0.50	0.62	-1.75	2.92
Mid distance bin	Discharge Year x x 2012 x treated	2.805*	1.41	1.99	0.05	-0.02	5.63
Mid distance bin	Discharge Year x x 2013 x treated	0.456	0.34	1.34	0.19	-0.23	1.14
Mid distance bin	Discharge Year x x 2014 x treated	0.202	0.27	0.76	0.45	-0.33	0.74
Mid distance bin	Discharge Year x x 2015 x treated	0.741**	0.30	2.44	0.02	0.13	1.35
Far distance bin	Discharge Year x x 2011 x treated	-3.920***	0.47	-8.37	0.00	-4.86	-2.98
Far distance bin	Discharge Year x x 2012 x treated	-2.205***	0.43	-5.14	0.00	-3.06	-1.35
Far distance bin	Discharge Year x x 2013 x treated	-2.595***	0.40	-6.44	0.00	-3.40	-1.79
Far distance bin	Discharge Year x x 2014 x treated	-0.998***	0.30	-3.30	0.00	-1.61	-0.39
Far distance bin	Discharge Year x x 2015 x treated	-0.062	0.37	-0.17	0.87	-0.81	0.68

Comment: This table reports separate formal pre-trend models for the near-, mid-, and far-distance Study 2 ZIP bins.
Significance codes: *** p < 0.01, ** p < 0.05, * p < 0.1.

Comment: This table reports separate formal pre-trend models for the near-, mid-, and far- distance Study 2 ZIP bins.

Overall, the parallel trends assumption is not strongly supported in the full Study 2 sample. While treatment and control ZIP codes exhibit broadly similar directional trends, differences in magnitude and slope persist across several specifications. Restricting the sample by geography and time improves comparability but does not fully eliminate pre-treatment differences. As a result, estimates from Study 2 should be interpreted with caution, particularly in light of potential violations of the identifying assumptions underlying the difference-in-differences framework.

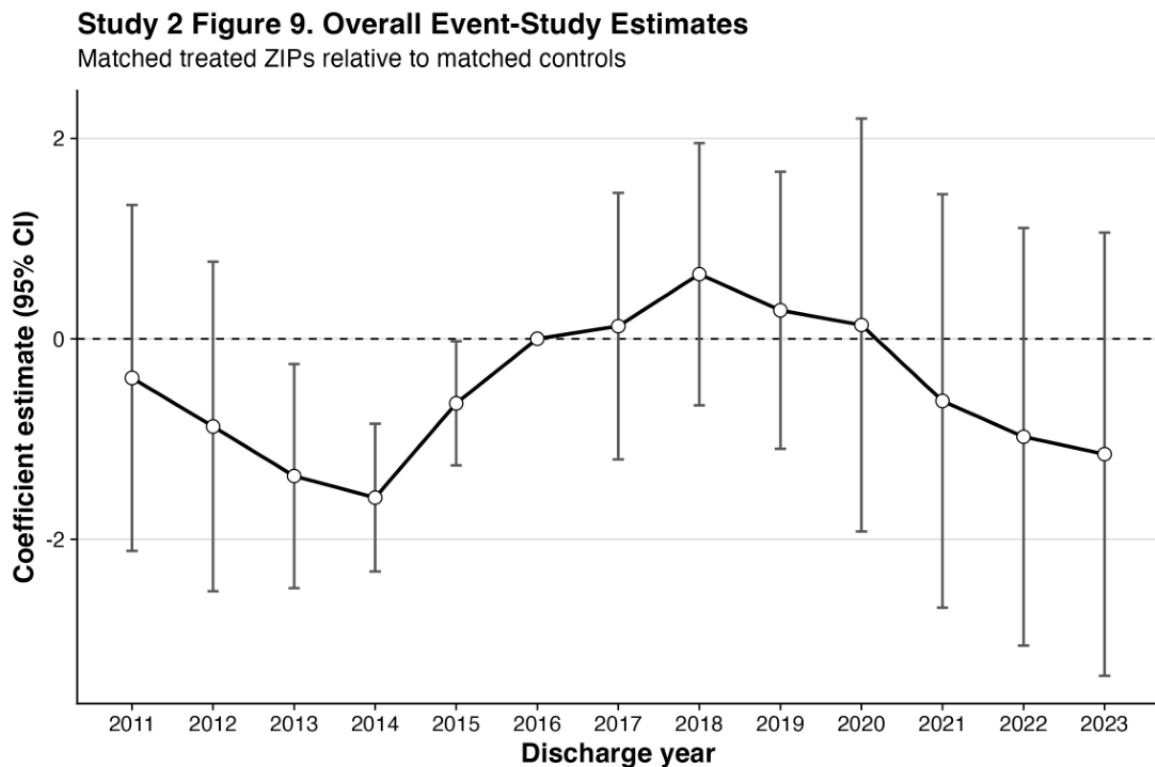
DiD Results

Overall

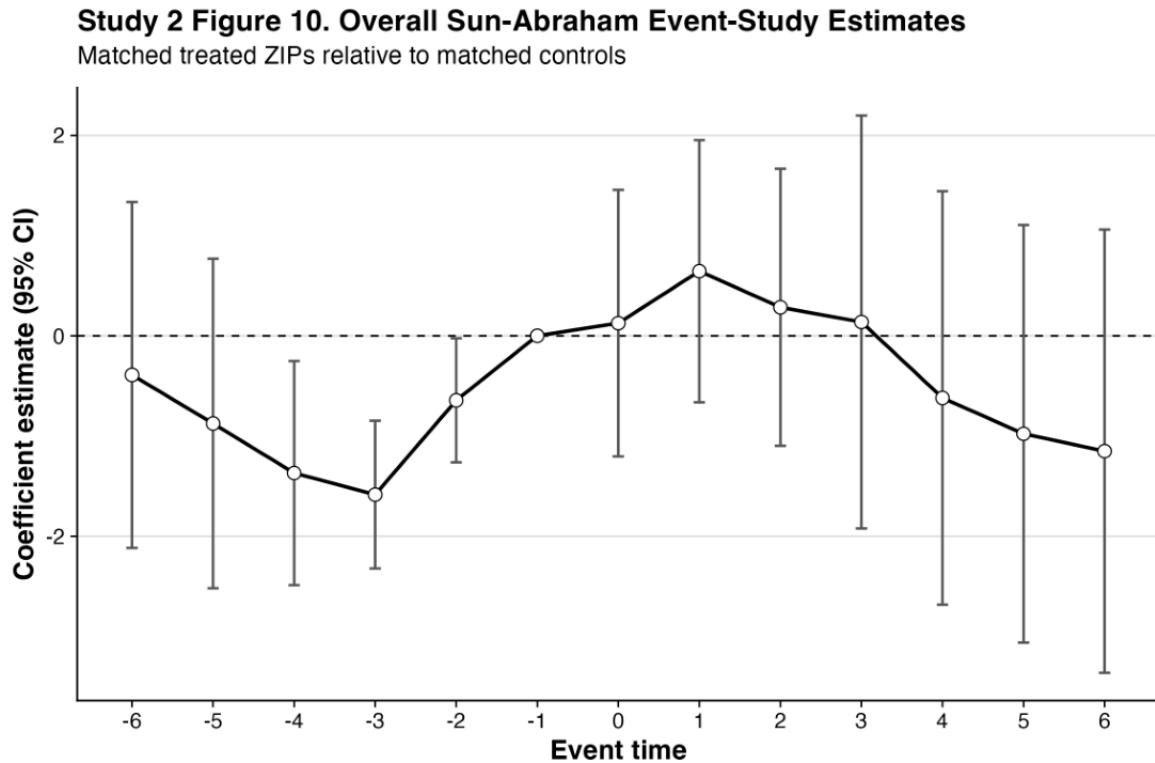
Across the matched sample, there was no statistically significant evidence that the Q-line expansion reduced or increased preventable emergency department (PPV) rates. Event study estimates show coefficients close to zero in most post-treatment years, with wide confidence intervals. Pre-treatment coefficients are not consistently flat, indicating imperfect parallel trends, though trends move in broadly similar directions.

Overall, the main specification does not provide statistically significant evidence of a change in potentially preventable emergency department visit rates; this is not evidence that the effect is exactly zero.

Event Study



Sun-Abraham Model



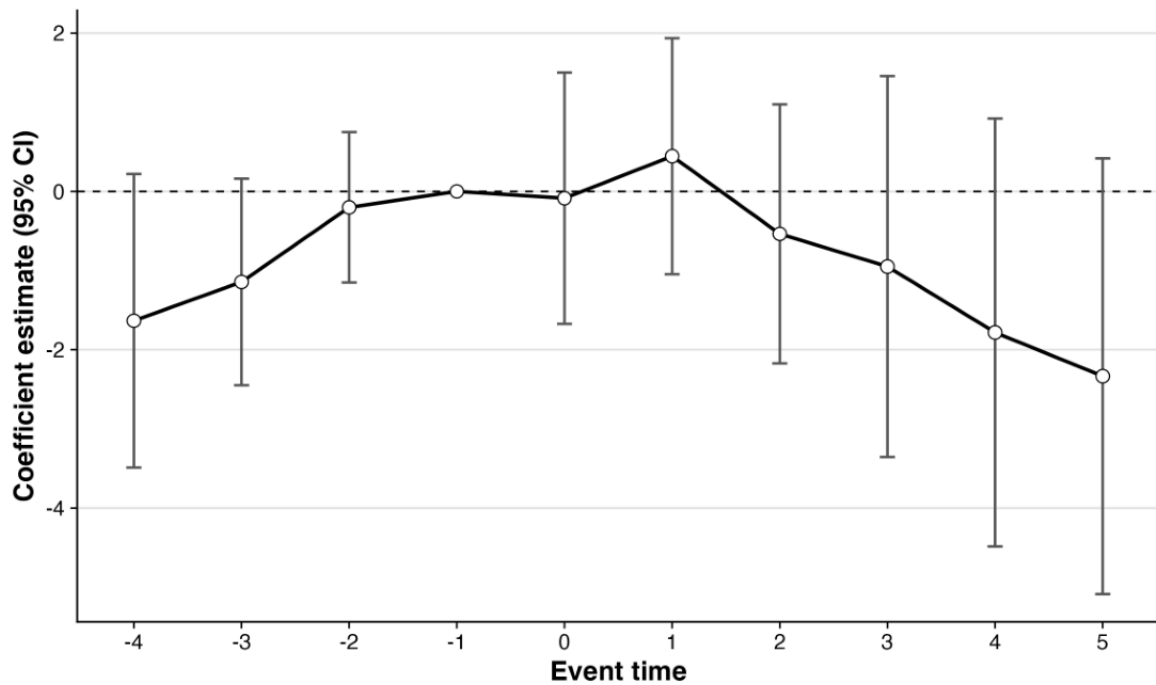
Heterogeneity between ZIP codes of different income levels and distances from the Upper East Side are examined in the following sections.

Income

For high-income ZIP codes, the estimated change in preventable ED visits in 2022 (event time +5) was negative but not statistically significant at the 5% level (-2.33 visits per 100 residents per year, $SE = 1.343$, $p = 0.093$), though I lack follow-up data to determine if this effect persists. For low-income ZIP codes, I found no significant effects of the station installation on PPV in low-income ZIPs. Though I cannot draw causal conclusions from this data due to the violation of the parallel trends assumption and risk of bias in the propensity matching process, it is possible that other factors had taken effect before the intervention (See the discussion section for further elaboration).

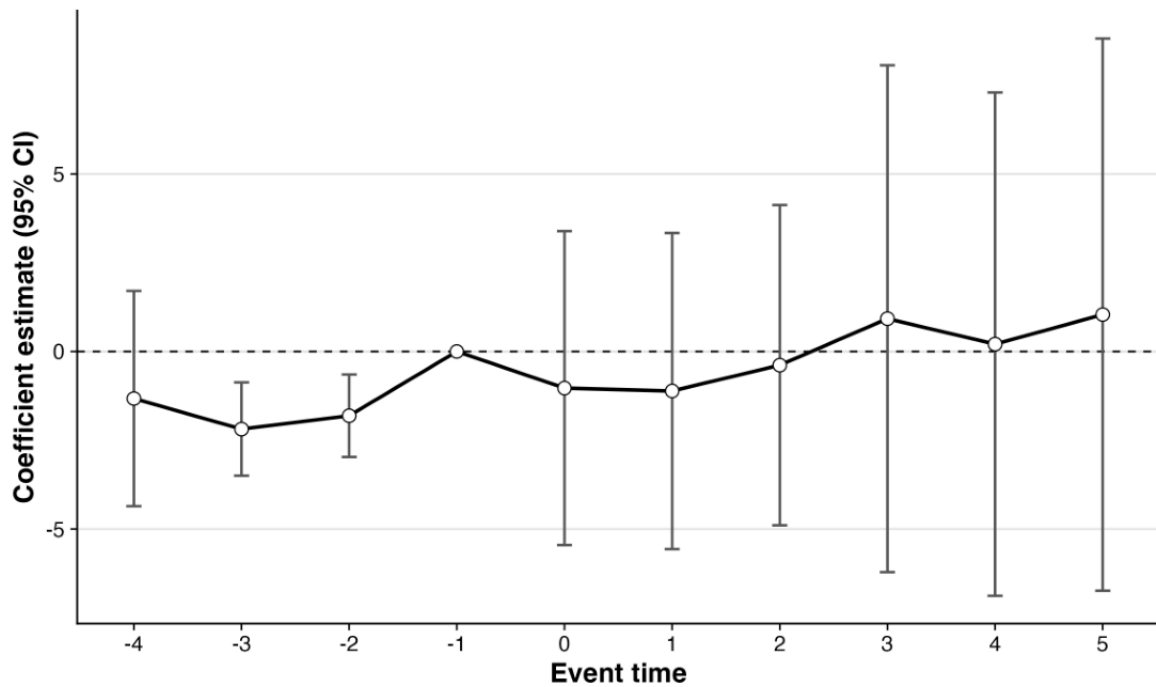
Study 2 Figure 11. High-Income ZIPs Sun-Abraham Event Study

Restricted to matched high-income ZIP codes, 2013-2022



Study 2 Figure 12. Low-Income ZIPs Sun-Abraham Event Study

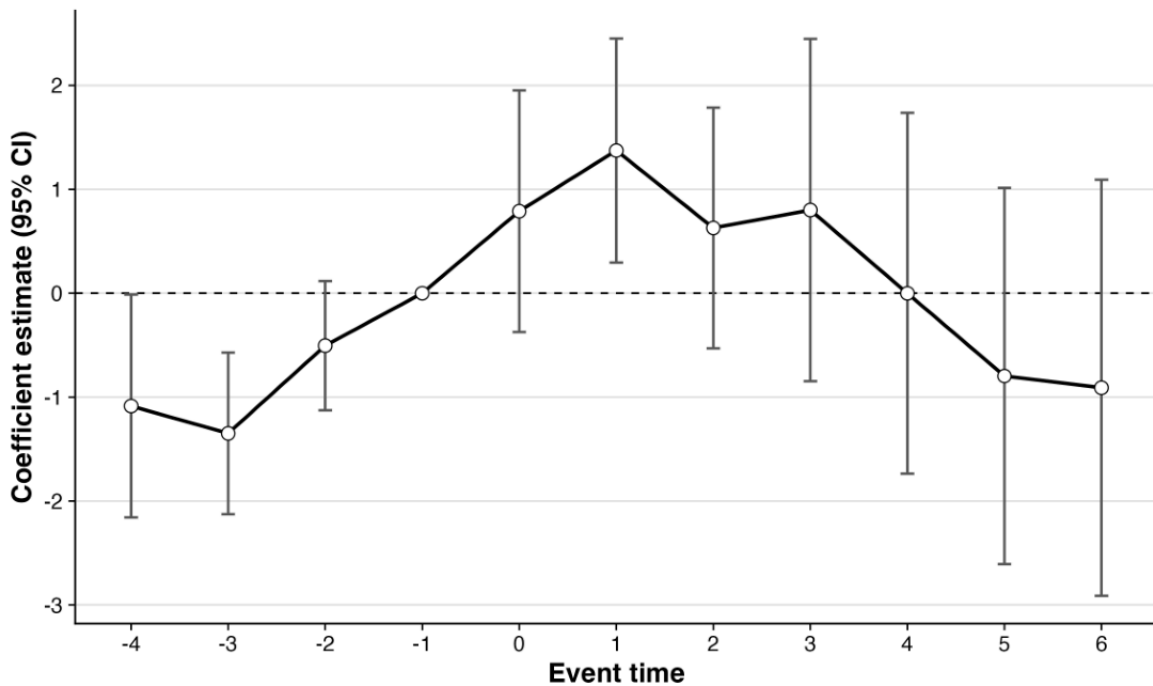
Restricted to matched low-income ZIP codes, 2013-2022



Distance

Study 2 Figure 14. Far ZIPs Sun-Abraham Event Study

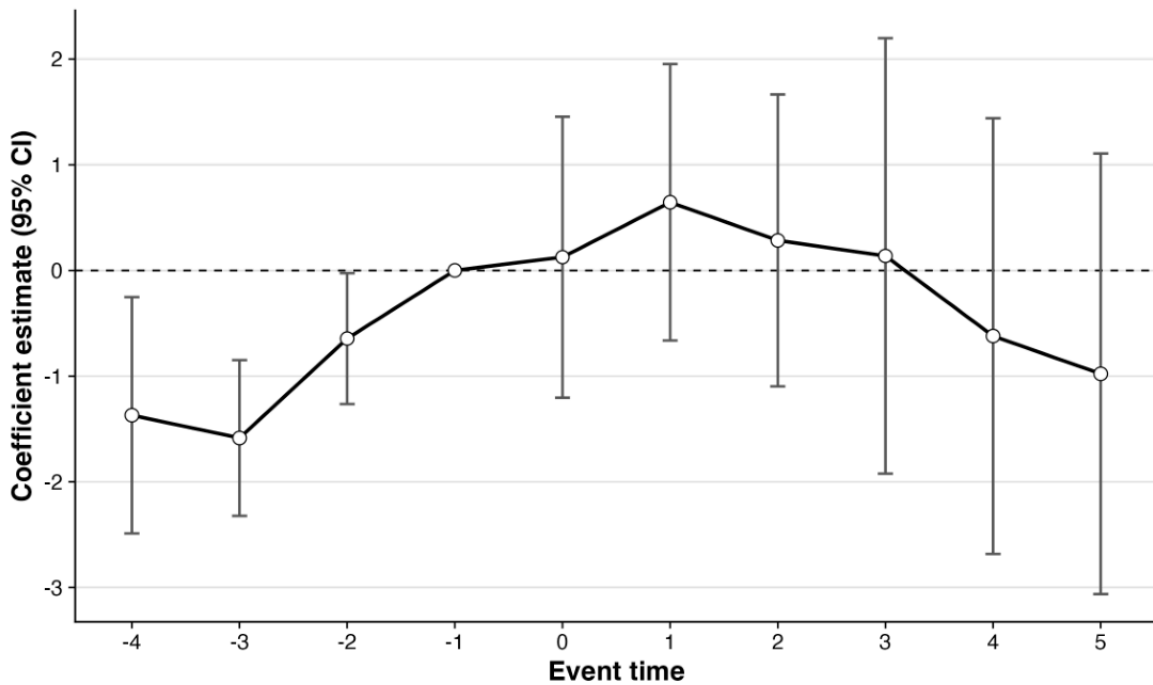
Restricted far-controls analytic sample



The ZIP codes farthest from the Upper East Side experienced a significant increase in risk-adjusted potentially preventable emergency department visit rates in 2018 (event time +1) (+ 1.37 visits per 100 residents per year, SE = .54, $p = .013$) relative to the 2016 reference year. However, these effects were not sustained in subsequent years. No statistically significant effects were observed for nearby ZIP codes (see corresponding figure).

Study 2 Figure 13. Draft 'Near ZIPs' Sun-Abraham Event Study

Recreates the thesis draft's 2013-2022 matched-sample restriction



Discussion

Study 1 examines the local effects of the subway expansion on potentially preventable emergency department visits in the area, with a positive but statistically nonsignificant estimated change in preventable ED visits post-treatment. The point estimate has the opposite sign to the initial hypothesis that increased transportation access would act as a mechanism to decrease potentially preventable outcomes in affected locales. However, the confidence intervals do not distinguish a reduction from an increase, and these estimates cannot identify the mechanisms underlying the observed pattern. It is also possible that the positive point estimate reflects changes driven by changes in air quality or environment created by the construction zone and subway traffic. A recent study by Azad et al. (2024) found that New York City commuters who ride the subway more often and for longer periods of time receive more exposure to harmful fine particulate matter (PM_{2.5}) pollution, disproportionately impacting low-income and outer borough New Yorkers. In this case, the addition

of new stations may have increased pollutants in the area. Moreover, it is possible that Upper East Side residents who might have once used the bus, a private vehicle, or walked to areas served by the Q-line may now choose to use the subway instead, increasing exposure to PM_{2.5} particles.

Study 2 similarly found largely statistically nonsignificant estimates across most specifications, with some heterogeneous effects across area income brackets, such that high-income Brooklyn and Manhattan ZIP codes had a negative estimate for ED visits that was not significant at the 5% level 5 years after the intervention, while low-income ZIPs showed no significant effects.

Several untested explanations warrant further investigation: (1) a single transit connection may be insufficient to materially change utilization patterns; (2) emergency and preventable hospitalizations may be driven by barriers other than transportation (financial access, insurance status); (3) the outcome may be too distal from the intervention, or (4) our ZIP code-level analysis may obscure individual-level variation in access and utilization. These findings can be considered alongside systematic reviews showing mixed evidence for transportation interventions on clinical outcomes (Solomon et al., 2020), and motivate further study of whether transportation access alone, without addressing concurrent barriers like financial constraints, may be insufficient to lower preventable emergency room rates.

For this study, I chose potentially preventable emergency department visit rates as the closest proxy for routine care access given limitations in data access and availability, based on the idea that patients who do not receive routine care will be more likely to be hospitalized for “preventable” worsening of pre-existing or new conditions, with the underlying assumption that (1) the Upper East Side offers superior outpatient medical care (particularly specialists), and that access to this care would improve condition management and early detection, and (2) that outer borough New Yorkers would increase their utilization of UES facilities given improved structural access (more convenient public transportation options). One of the greatest limitations of this paper lies in the absence of

patient-level outpatient utilization and destination data. Future studies should compare UES facility visitation rates before and after the Second Avenue opening, as it is yet unclear whether those facilities actually saw an increase in patients following the intervention.

Regarding the exploratory differences between high- and low-income ZIP codes, it is well known that medical costs are a huge barrier to access in the US; a plausible explanation for the negative point estimate in high-income ZIPs is that the proposed mechanism (access improves utilization, which improves outcomes) may operate differently across groups for whom convenience is the highest barrier: Increased physical access means little for patients who are unable to afford care regardless of location. Indeed, a 2025 report from the NYC comptroller's office found rising healthcare costs and financial distress, even among insured residents. Notably, prescription drug costs were a significant concern, contributing to high out-of-pocket costs and premiums for consumers, as were ongoing labor shortages in the medical field, particularly in primary care (Office of the New York City Comptroller, 2025). Besides the obvious health impact of being unable to take medication as prescribed due to rising costs, a decline in primary-care professionals is particularly concerning with regard to preventive care. If patients are unable to be seen due to long wait times and appointment unavailability, then improving transportation alone may not resolve these barriers.

Next, the Covid-19 pandemic is an obvious confound in the results. Though in study 1 we did see a small increase in hospitalizations long before the pandemic arrived in the US, any lagged results are impossible to differentiate. The small lagged decrease in PPV seen in wealthy neighborhoods could potentially be a result of improved practices following the pandemic, such as improved masking practices that reduce disease transmission, or decreased outdoor activity that results in fewer accidents. Future research should examine the complex interplay between shifting post-pandemic cultural practices, economic concerns, and public transportation.

Implications for policy

These analyses do not establish that the Q-line expansion reduced potentially preventable ED visit rates. Neither do they establish that transportation access is unimportant to health outcomes in New York City. The findings in the literature on this topic are mixed: Studies by Blazer et al. (1995) and Skinner et al. (2007) found no significant difference between rural and urban adults when it came to transportation barriers and the deferral of medical care when socioeconomic status was controlled for, while other studies found that rural adults were more burdened by travel distance and by lack of access to transportation than their urban counterparts (Blazer et al. (1995) and Skinner et al. (2007; as cited in Syed et al., 2013). Given this disagreement in the literature, and the relatively unique environment of New York City, it is unclear if the present study's results are generalizable across American populations. Future research comparing New York City to other American cities is needed.

One possible explanation is that financial constraints may be particularly salient for NYC patients. Along with rising health care costs (Office of the New York City Comptroller, 2025), New Yorkers also overwhelmingly struggle to afford city life; the March 2026 Cost of Living Report from the NYC Mayor's Office of Equity & Racial Justice reports that 62% of New Yorkers were unable to meet the true cost of living in the city in 2022 (NYC Mayor's Office of Equity & Racial Justice, 2026). These affordability concerns merit attention, but the present study does not compare affordability interventions with transportation investments and cannot establish which should receive priority.

Limitations

This dataset is the best available for this study because it includes aggregations of real clinical data, rather than relying on self-reported outcomes. At the same time, there are several key limitations that impede our ability to draw causal claims.

A central limitation of this project is the Study 2 violations of the parallel trends assumption. Meeting the parallel trends assumption is critical for DiD, meaning this violation undermines a causal interpretation of Study 2 estimates. Similarly, the risk of bias in the Study 2 propensity score matching procedure indicates that any results should be interpreted as strictly exploratory.

Similarly, the SPARCS dataset also has several key limitations. Because it provides ZIP code level data, it is impossible to disentangle individual-level factors that may influence their healthcare utilization patterns; moreover, our analyses implicitly assume all residents of a given ZIP code share the same distance to and from their ZIP code's subway station, which is an assumption that does not reflect reality.

Another limitation of this study is the use of only one outcome variable (risk-adjusted preventable ED visits) as a proxy for healthcare access, under the assumption that an increase in “preventable” hospitalizations, as determined by SPARCS, would result from a decrease or lack of access to routine care (eg. a hospitalization that could have been prevented by access to earlier care). However, this captures only one dimension of healthcare access. Future research should investigate other types of clinics and patient outcomes. For example, chronic condition instances and management should be directly linked to subway access under the current model as both an outcome and a moderator. It is likely that certain populations benefit more from increasing convenience than others.

Finally, the time period examined in this study is intercepted by the 2020 COVID-19 pandemic, making it impossible to disentangle possible delayed effects from the subway intervention and the pandemic.

Future Directions

Study 1 finds a small positive but statistically nonsignificant estimate for potentially preventable emergency department visits following the subway expansion, while Study 2 finds largely null or mixed patterns across the Q-line corridor, even when broken down by income and distance. Taken together, these two studies do not provide evidence to support my hypothesis that PPV would decrease following improved transportation access through increased access to routine medical care at Upper East Side medical centers. Study 2 should additionally be interpreted as exploratory given its methodological limitations. However, the results also do not establish that the expansion increased PPV or had no effect, nor are we able to disentangle the individual mechanisms that may underlie the observed patterns.

Future analysis should employ individual level data. Our heterogeneous results indicate that intervention impact may differ across populations. It is likely that elderly and chronically ill populations may benefit more from improved public transportation

Future research should also compare actual total hospital/outpatient utilization rates to analyze total facility utilization. Data should also include utilization of outpatient clinics and primary care practices, as those may provide more direct measures of preventative care utilization. Finally, future research should compare other instances of increased or decreased transportation access. For example, the 2005 MTA strike might be another natural experiment and would allow for comparisons of outcomes across political and environmental contexts.

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Appendix

A. Derived variables, formulae, and ACS source tables

Derived variable	How it was derived	ACS source variable(s)	ACS table
income	Median household income, used directly from ACS estimate	B19013_001	B19013
poverty_rate	Number below poverty divided by poverty universe/denominator	B17001_002 / B17001_001	B17001
total_pop	Total population from race/ethnicity table, used directly	B03002_001	B03002
pct_white	Non-Hispanic White population divided by total population	B03002_003 / B03002_001	B03002
pct_black	Non-Hispanic Black population divided by total population	B03002_004 / B03002_001	B03002
pct_hispanic	Hispanic population divided by total population	B03002_012 / B03002_001	B03002
college_educated	BA + graduate degree counts for men and women divided by education universe	(B15002_015 + B15002_016 + B15002_017 + B15002_018 + B15002_032 + B15002_033 + B15002_034 + B15002_035) / B15002_001	B15002
income_pre	Mean income across pre-treatment years	mean income, 2011–2016	Derived from B19013
income_post	Mean income across post-treatment years	mean income, 2017–2022 in code logic	Derived from B19013
college_pre	Mean college-educated share across pre-treatment years	mean college_educated, 2011–2016	Derived from B15002
college_post	Mean college-educated share across post-treatment years	mean college_educated, 2017–2022 in code logic	Derived from B15002
poverty_pre	Mean poverty rate across pre-treatment years	mean poverty_rate, 2011–2016	Derived from B17001
poverty_post	Mean poverty rate across post-treatment years	mean poverty_rate, 2017–2022 in code logic	Derived from B17001
d_income	Change in average income from pre to post period	income_post - income_pre	Derived from B19013
d_college	Change in average college-educated share from pre to post period	college_post - college_pre	Derived from B15002

Derived variable	How it was derived	ACS source variable(s)	ACS table
d_poverty	Change in average poverty rate from pre to post period	poverty_post - poverty_pre	Derived from B17001
rel_income	ZIP-level income change minus NYC-wide income change	d_income - d_income_nyc	Derived from B19013
rel_college	ZIP-level education change minus NYC-wide education change	d_college - d_college_nyc	Derived from B15002
rel_poverty	ZIP-level poverty change minus NYC-wide poverty change	d_poverty - d_poverty_nyc	Derived from B17001
gentrification_index	Standardized relative income change + standardized relative education change – standardized relative poverty change	scale(rel_income) + scale(rel_college) - scale(rel_poverty)	Composite derived from B19013, B15002, and B17001

B. ZIP Codes Defined as Proximate to the Q-Train

Manhattan

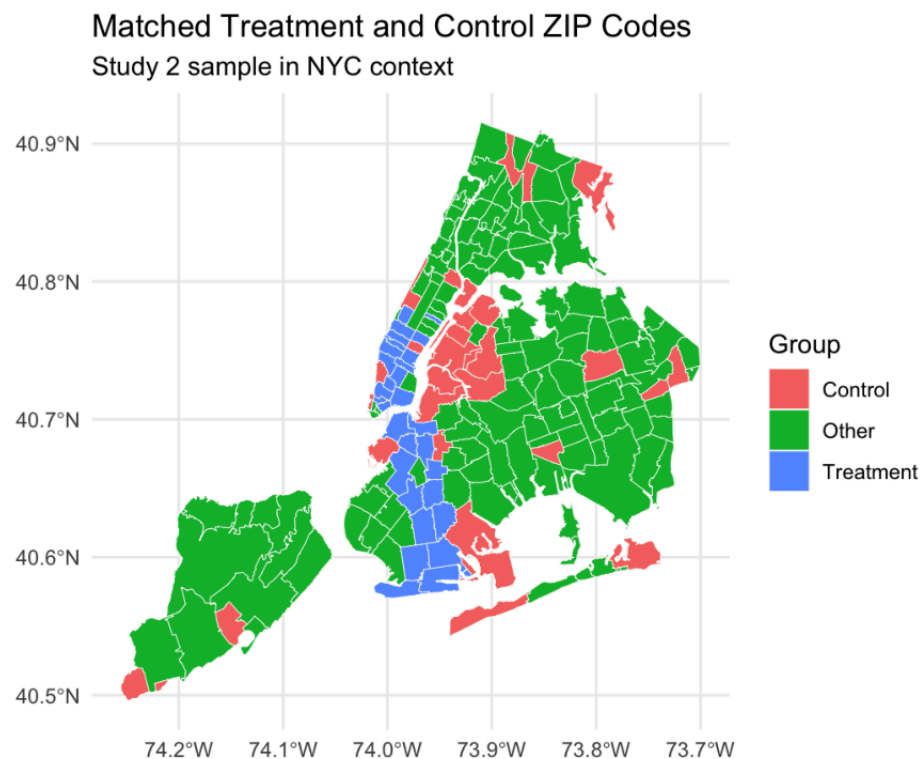
10001 — Chelsea / Hudson Yards
 10002 — Lower East Side
 10003 — East Village / Union Square
 10007 — Tribeca / Civic Center
 10010 — Gramercy / Flatiron
 10011 — Chelsea / Meatpacking District
 10012 — SoHo / NoHo
 10013 — Tribeca / Chinatown
 10016 — Murray Hill / Kips Bay
 10018 — Garment District / Midtown South
 10019 — Midtown West
 10021 — Upper East Side
 10022 — Midtown East
 10023 — Upper West Side
 10028 — Upper East Side
 10029 — East Harlem
 10036 — Midtown / Times Square
 10038 — Financial District
 10065 — Upper East Side (Lenox Hill)
 10075 — Upper East Side
 10128 — Upper East Side (Carnegie Hill)

Brooklyn

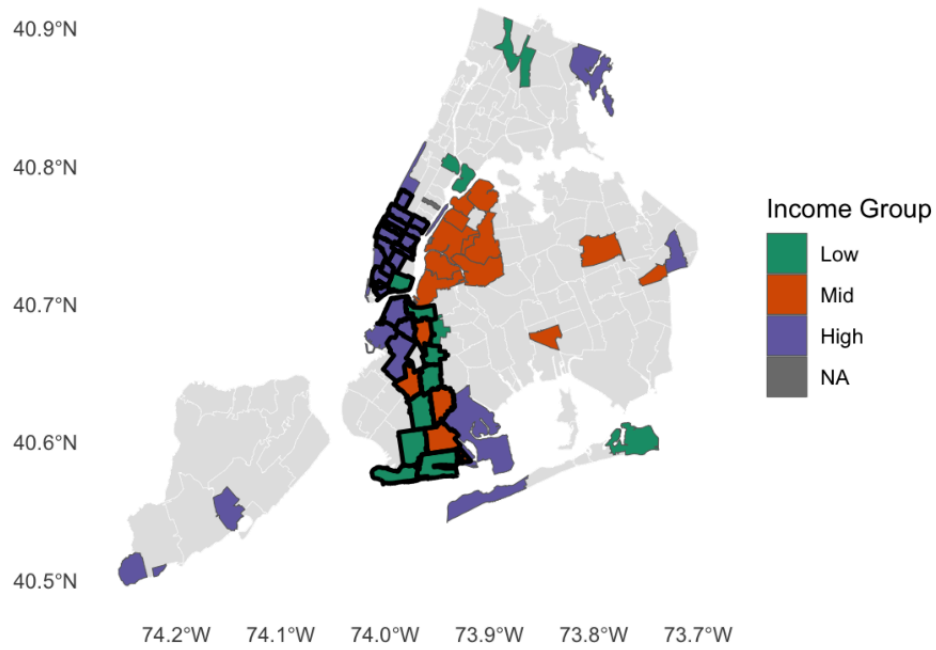
11201 — Brooklyn Heights / Downtown Brooklyn
 11205 — Clinton Hill / Fort Greene
 11210 — Flatbush / Midwood

11215 — Park Slope / Gowanus
 11217 — Boerum Hill / Fort Greene
 11218 — Kensington / Windsor Terrace
 11223 — Gravesend
 11224 — Coney Island
 11225 — Crown Heights
 11226 — Flatbush
 11229 — Sheepshead Bay
 11230 — Midwood
 11235 — Brighton Beach
 11238 — Prospect Heights / Clinton Hill

C. Study 2 Treatment and control ZIPs were mapped using spatial shapefiles. ZIP codes were labeled as treated, control, or other and visualized geographically.



Study 2 ZIP Codes by Income Group
Treatment ZIPs outlined; controls shown without outline



Study 2 ZIP Codes by Distance Group
Treatment ZIPs outlined; controls shown without outline

